Reasoning Your Way Out of Anxiety? A Critical Review of Cognitive Restructuring, Executive Function and Adult ADHD

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Abstract

Cognitive behavioral therapy (CBT) is the best-studied psychotherapy for anxiety, yet roughly half of treated patients do not respond, and the mechanism usually offered for its cognitive component—that patients change feelings by examining and replacing automatic thoughts—has seldom been tested against the cognitive demands it places on the patient. This review makes one argument. Cognitive restructuring is, mechanically, an executive-function task (working memory, inhibition, shifting). Anxiety and acute stress impair the prefrontal systems that support these functions, and attention-deficit/hyperactivity disorder (ADHD) is characterised by executive weaknesses before anxiety arrives. We therefore ask whether the "argue with your thoughts" component carries the benefit of CBT, or whether exposure, behavioural skills and repeated practice do. We (i) specify the executive demands of restructuring, (ii) review evidence on stress, anxiety and prefrontal function, (iii) review the efficacy evidence for CBT in anxiety and in adult ADHD, including the alternatives (mindfulness-based programmes, acceptance and commitment therapy, medication), (iv) review the sparse direct evidence on whether executive function moderates CBT outcome, and (v) steelman the case against the argument. The evidence supports a narrow claim and does not support a broad one: restructuring-heavy CBT is effective on average, but its cognitive mechanism is under-tested, no adequately powered trial has tested executive function as a moderator in working-age

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adults with ADHD and anxiety, and the best-supported active ingredients are behavioural. We close with preregistrable predictions and a research agenda. **This is draft 1; claims marked [CITE] await sources.**

Keywords: cognitive behavioral therapy, cognitive restructuring, executive function, anxiety, ADHD, mindfulness, critical review

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1. Introduction

Cognitive behavioral therapy (CBT) is among the most heavily tested psychotherapies. A meta-analysis of 41 placebo-controlled trials in anxiety and related disorders (total $N = 2{,}843$) found that CBT outperformed placebo on target symptoms with a moderate effect (Hedges' $g = 0.56$) (Carpenter et al., 2018). The same literature also shows a ceiling: across 87 studies, only about half of patients respond to CBT for anxiety disorders, and the figure depends heavily on how response is defined (Loerinc et al., 2015). A therapy can work on average and still leave open *why* it works and for *whom*.

The textbook account of CBT's cognitive component is a top-down loop. A situation triggers an automatic thought ("I am going to fail this talk"); the patient records it, names the distortion, weighs evidence for and against, and generates a more balanced thought; the feeling is expected to follow. This review asks a narrow question about that loop: *does the mechanism ask the patient to use a capacity that anxiety, and in some patients ADHD, have already degraded?* If so, the cognitive component should work best when practiced calm, should depend on baseline executive capacity, and should contribute less to outcome than behavioural elements that do not demand effortful reasoning.

Scope and method. This is a critical (theoretical) review. It does not claim that CBT is ineffective, and it does not present a new systematic search of all CBT trials. It selects the literatures needed to evaluate one mechanistic argument, states what each can and cannot show, and lists what we could not find. All citations were checked against the publisher record (DOI resolved through Crossref) and, for numerical claims, against the abstract or full text; the search and verification record is in the repository (`deep-research/`). Claims we could not source are marked in the text. Popular statements of the argument, notably by Judson Brewer (Brewer, 2016, 2021, 2026), are treated as claims to be evaluated, not as evidence.

Contributions. (1) An explicit decomposition of cognitive restructuring into executive demands (Section 2). (2) A review of why acute stress and anxiety should degrade those demands, with caveats on the animal-heavy mechanistic literature (Section 3). (3) A fair assessment of the efficacy evidence for CBT in anxiety and adult ADHD, and of the alternatives (Sections 5–7). (4) A summary of the thin direct evidence on executive function as a moderator of CBT outcome (Section 8). (5) Preregistrable predictions (Section 10).

2. What cognitive restructuring asks of the patient

Consider the standard thought-record exercise. The patient must (a) *notice* an automatic thought while it is occurring, (b) hold the thought, the supporting evidence, the contradicting evidence and a candidate alternative in mind at once, (c) suppress the pull of the automatic thought while evaluating it, and (d) shift from being inside the feeling to observing it. Mapped onto the best-known model of executive function, these are three separable components: updating and holding information (working memory), inhibiting a prepotent response, and shifting between mental sets (Miyake et al., 2000). Executive function is, to a first approximation, supported by prefrontal networks.

Two clarifications keep this claim honest. First, it is a task analysis, not an empirical result: the argument needs the demands to be *real* in practice, and clinicians teach scaffolds (worksheets, a therapist who prompts the evidence, repetition) that lower the load. Second, executive function is not a single resource; the argument concerns the three components above, not a general "intelligence" required for therapy. We return to both points in Section 9.

3. Stress, anxiety and the prefrontal executive system

3.1. Mechanistic neuroscience

The most developed account is due to Arnsten. Her review of the molecular signalling underlying stress effects describes the prefrontal cortex (PFC) as the most evolved brain region and also the one most sensitive to the detrimental effects of stress exposure (Arnsten, 2009). The proposed mechanism is an inverted-U in catecholamine signalling: moderate norepinephrine and dopamine levels support prefrontal network firing and working memory, while the high levels of uncontrollable stress weaken them, and strengthen

amygdala and striatal habit circuitry. A later review summarises the result as stress rapidly flipping the brain from reflective to reflexive control of behaviour (Arnsten, 2015). Evolutionarily this is intelligible: a threat favours fast reflexes over deliberation.

Caveats. Much of the mechanistic evidence comes from rodents and non-human primates. "Offline" is a popularisation of *impaired*: the data support graded, dose-dependent impairment, not an on/off switch. Mild or controllable stress can leave executive function intact or improved. Brewer (2021) and Brewer (2016) state the strong version for a lay audience (anxiety shuts the PFC down and, with it, willpower and reasoning). We read these as clinician assertions that the primary literature supports only in graded form. [CITE: human acute-stress meta-analyses of working memory, inhibition and shifting; reappraisal under stress]

3.2. Cognitive psychology of anxiety

Attentional Control Theory proposes that anxiety impairs the inhibition and shifting functions of executive control, so that anxious individuals are more captured by threat and worse at redirecting attention (Eysenck et al., 2007). An important nuance is that the theory predicts *efficiency* (effort and time) to suffer more than *effectiveness* (accuracy), because anxious people compensate with extra effort. That compensation is expensive for exactly the kind of person who has little executive reserve, which is the point of the next section.

4. Executive function in ADHD

ADHD is a heterogeneous disorder in which executive difficulties are common but not universal. A meta-analysis of 83 studies found that groups with ADHD underperform controls on every executive task class examined, most strongly on response inhibition, vigilance, working memory and planning, with medium effect sizes (0.46–0.69) (Willcutt et al., 2005). The same authors caution that executive weaknesses are neither necessary nor sufficient to cause all cases of ADHD, and that fewer than half of the (mostly child) samples show impairment on any single executive test (Willcutt et al., 2005). Any argument that treats ADHD as uniformly executive-impaired overstates the evidence.

Even so, in the subgroup with executive weaknesses, the conjunction is troubling for restructuring-heavy therapy: baseline executive capacity is low,

anxiety lowers it further, and, per Attentional Control Theory, the compensatory effort that anxious people deploy is itself an executive resource (Eysenck et al., 2007). The comorbidity of anxiety and ADHD in adults makes this a clinically common situation. [CITE: prevalence of anxiety in adult ADHD; anxiety-disorder CBT outcome when ADHD is comorbid]

5. What the efficacy evidence shows

5.1. CBT for anxiety

The strongest case for CBT is large and should be stated plainly. In placebo-controlled trials, CBT beats placebo on target symptoms with a moderate effect ($g = 0.56$ across 41 trials, $N = 2{,}843$), and roughly triples the odds of responding (Carpenter et al., 2018). This evidence is the reason any critique has to explain why CBT works at all. The limits are also clear: response rates average about half of patients, and they vary with the criteria used (Loerinc et al., 2015). We found no evidence in this literature that speaks to the *mechanism*: trials randomise to a package, not to the cognitive-restructuring component. [CITE: effect-size trend over publication year; waitlist versus active control; risk of bias; component and dismantling trials]

5.2. CBT for adult ADHD

In a randomised trial of 86 medication-treated adults with persistent ADHD symptoms, CBT outperformed an attention-matched control of relaxation with educational support: 53% versus 23% were responders on clinician global ratings (Safren et al., 2010). Two features matter for our argument. Every participant was already medicated, and the programme was skills-heavy (organising and planning, managing distractibility, procrastination) with adaptive thinking as one module among several. The Cochrane review of CBT for adult ADHD concludes that the evidence is of low quality, with short-term benefit possible but unclear (Lopez et al., 2018). These findings are compatible with a reading in which the active ingredient is concrete skills and habit-building, not reappraisal, but they do not isolate the component.

6. Where the active ingredients may lie

A defender of CBT has an obvious reply: patients do not restructure during a panic; they practice when calm, so that the response becomes automatic. We think this is the strongest objection to the executive-function

argument, and we accept its force. It also changes what is being claimed. If repeated rehearsal during calm periods does the work, the active ingredient is practice and habit formation, not in-the-moment reasoning, and the relevant question is how much executive effort each rehearsal costs.

A second observation points the same way. Exposure, which is central to CBT for anxiety, is behavioural: the patient approaches the feared stimulus and learns, from experience, that the feared outcome does not occur. Its logic resembles extinction learning more than debate. If most of the benefit of the CBT package comes from exposure and skills practice, the argument against *restructuring* as the key ingredient is strengthened without any claim that CBT does not work. [CITE: dismantling trials (Jacobson et al.), component meta-analyses, inhibitory-learning accounts of exposure, homework compliance meta-analyses]

7. Alternatives and their evidence

7.1. Mindfulness and acceptance-based treatments

The alternative that follows from the argument is to train attention and acceptance directly instead of disputing thoughts. Brewer frames anxiety as a habit loop (trigger, worry, a feeling of control) and proposes noticing, curiosity about the reward value of worry, and a "bigger, better offer" in place of argument (Brewer, 2021). Evidence for this specific programme is early: in a randomised trial of an app-based programme for generalised anxiety ($n = 65$), anxiety scores fell by 67% versus 14% under usual care, with mediation analyses implicating reduced worry (Roy et al., 2021). The trial is small, uses self-report, compares with usual care rather than CBT, and has a developer among the authors.

Against evidence-based comparators, mindfulness-based programmes do not outperform them: a meta-analysis found essentially no difference (Hedges' $g \approx 0.00$) against evidence-based treatments, often CBT (Goldberg et al., 2018), and acceptance and commitment therapy (ACT) versus CBT shows the same pattern (A-Tjak et al., 2014). In a trial of 276 adults with anxiety disorders, an eight-week mindfulness-based stress reduction course was non-inferior to escitalopram, with fewer discontinuations for adverse events (Hoge et al., 2023). That is parity with a first-line drug, not superiority to CBT. In adult ADHD, mindfulness-based cognitive therapy added to usual care showed a benefit ($d = 0.41$; Janssen et al., 2018), while a later meta-analysis found only small effects against active controls (Oliva et al., 2021).

7.2. Medication

In a network meta-analysis of ADHD medications, stimulants outperformed placebo in adults, with standardised effects of about 0.8 (amphetamines) and 0.5 (methylphenidate) on clinician ratings, over short follow-up of about 12 weeks (Cortese et al., 2018). If baseline executive capacity is the bottleneck, raising it pharmacologically before layering skills is mechanistically coherent, but this review does not test that combination.

8. Does executive function moderate CBT outcome?

This is the question on which the argument stands or falls, and the direct evidence is thin. In a pilot study of anxious older adults, participants whose executive function remained low did not respond to CBT, while those whose executive function improved did (Mohlman and Gorman, 2005). In another sample, low cognitive flexibility did not predict overall CBT outcome but did predict less acquisition of cognitive-restructuring skill specifically (Johnco et al., 2014); the second result is the closest published evidence to the claim made here. Both studies are in older adults, are small, and cut in different directions on overall outcome. We did not find an adequately powered study of working-age adults with ADHD and anxiety. [CITE: executive-function moderators in younger adults, youth, OCD and depression; Pimontel-type meta-analyses]

9. The strongest objections

1. **CBT has a large evidence base.** Yes; Section 5. The argument is about mechanism and moderators, not average efficacy.
2. **CBT for adult ADHD works.** A trial shows benefit over a matched control (Safren et al., 2010), but in medicated adults, with a skills-heavy protocol, and the Cochrane review grades the evidence as low quality (Lopez et al., 2018).
3. **Patients practice when calm.** Accepted in Section 6; it moves the active ingredient to practice.
4. **Exposure is behavioural.** Likewise; it strengthens the argument against restructuring specifically.
5. **Mindfulness is not better.** True (Section 7); the argument does not require superiority, only that restructuring is not uniquely necessary.

6. **The neuroscience is about acute stress.** Correct. The stress literature does not directly test whether weaker executive function predicts worse CBT outcome. Section 8 reviews the sparse direct evidence.
7. **Therapists scaffold the executive load.** Plausible and untested. A therapist who generates the evidence list for the patient lowers the demand and could erase the moderator effect.

Limitations of this review. It is a narrative critical review, written by a non-clinician; search was targeted rather than exhaustive in this draft; and the claims about mechanism are theoretical.

10. Predictions and a research agenda

The argument yields predictions that could fail:

1. **P1 (moderation).** Baseline executive function, measured before treatment, predicts the acquisition of restructuring skill more than the acquisition of exposure-based skills.
2. **P2 (stress-dependence).** Cognitive reappraisal efficacy falls with stress intensity and with lower executive capacity, so the gap between "calm" and "in-the-moment" use is larger in low-executive-function samples.
3. **P3 (component).** In adult ADHD with anxiety, a behaviour-and-exposure-only protocol is not inferior to the same protocol plus restructuring.
4. **P4 (ADHD).** ADHD status predicts less benefit from the cognitive module specifically, not from the skills modules.

Evidence that would refute the argument: an adequately powered study finding no moderation by executive function; component trials showing that restructuring adds substantial benefit; or meta-regression showing that executive-function-heavy samples respond as well. A secondary analysis of existing trial data, or an individual-participant meta-analysis, is the cheapest next step (see `experiments/`).

11. Conclusion

Restructuring-heavy CBT is effective on average. What is not established is that the cognitive component is the active ingredient, or that it works

for people whose executive function is low. The best-supported claim is narrow: the argue-with-your-thoughts component is the part least supported mechanistically and most exposed to executive limits, while exposure, skills and practice look like the parts that carry the benefit.

Declarations

Funding: none. **Competing interests:** none declared. **Author note:** this is a theoretical critical review by a non-clinician; it is not medical advice. **Data and materials:** the search record, verified bibliography and analysis logs are in the public repository accompanying this paper.

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